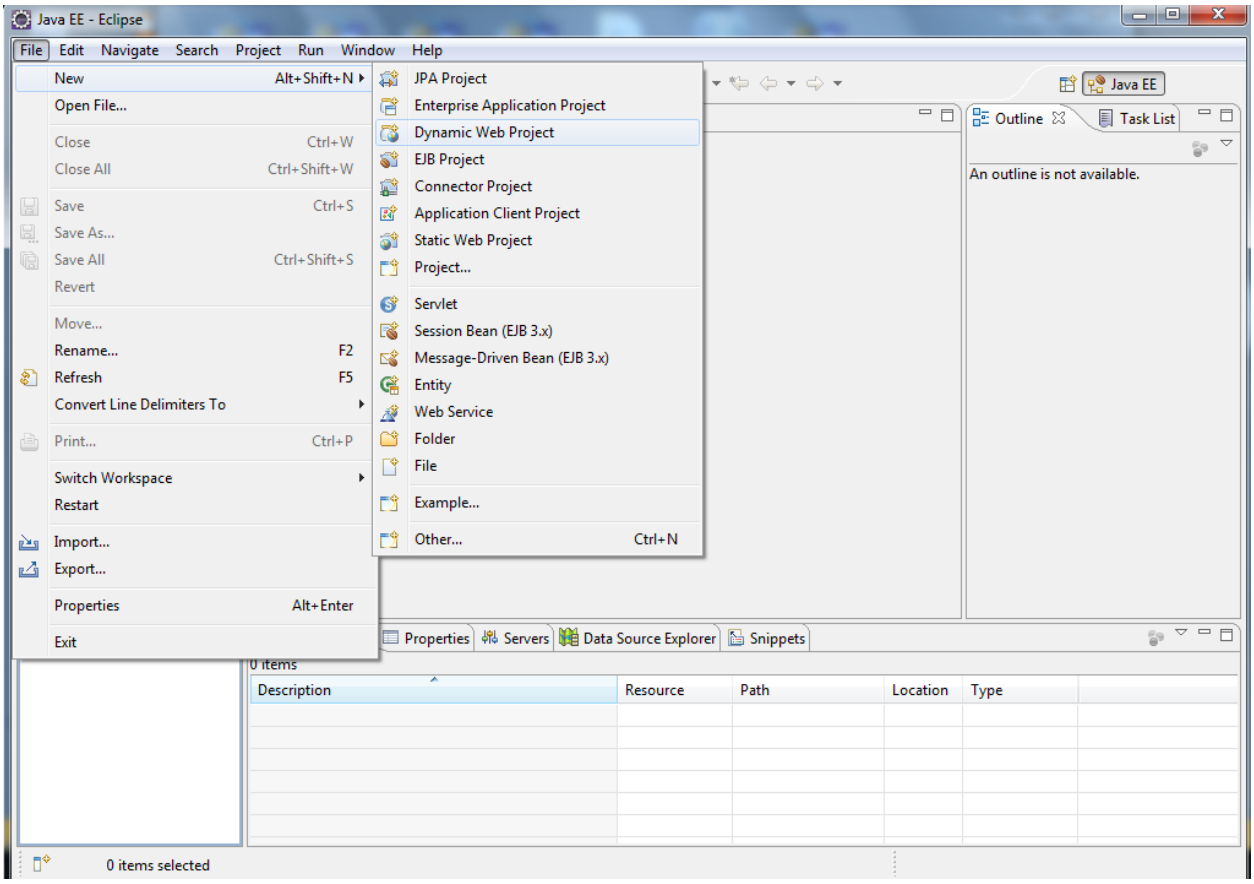
# Servlet - Session Management

## Problem Statement

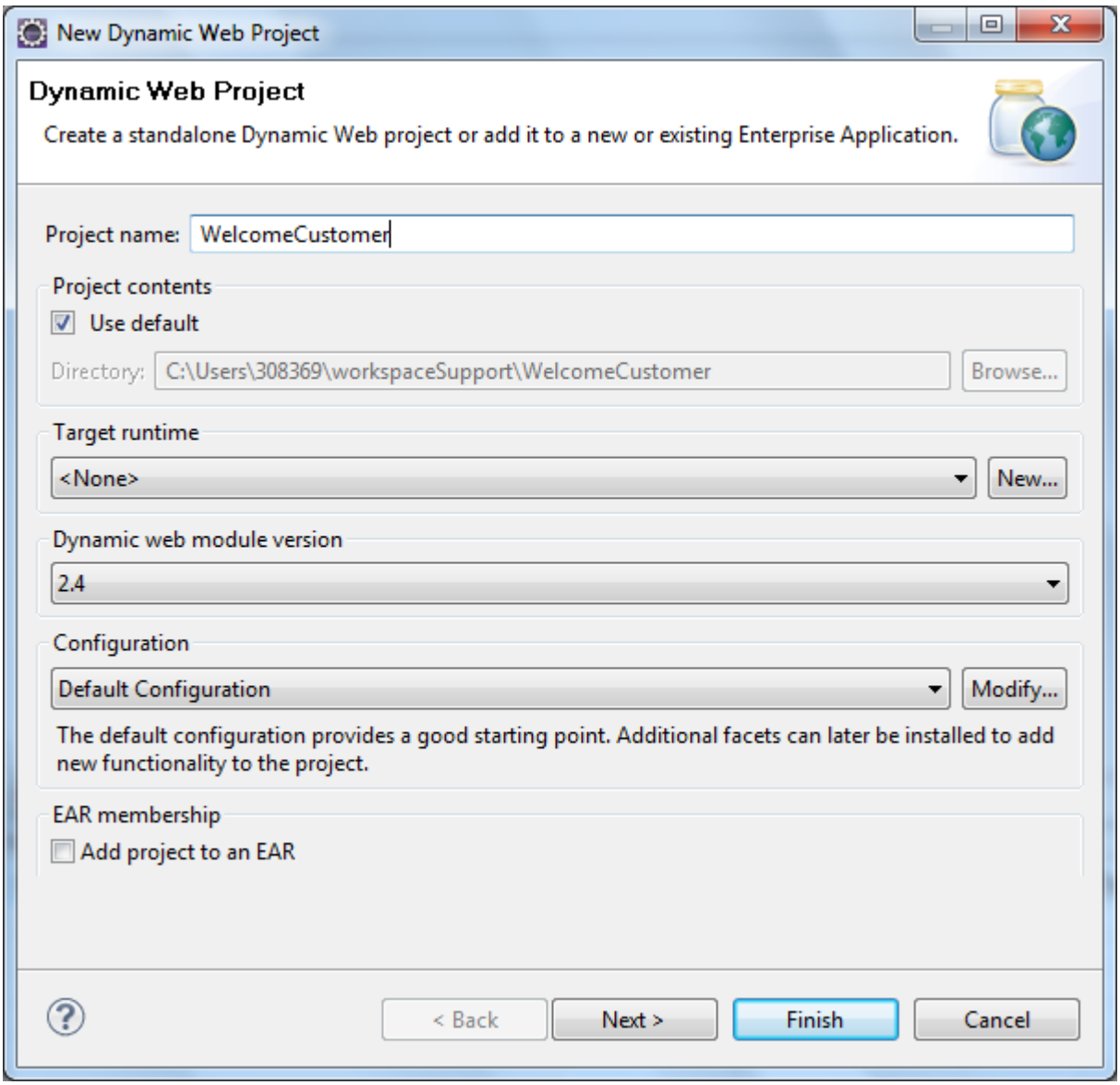
Develop a simple java web application to implement the session management functionality. The sytem should get the customer name and store it session. When the user registers, the welcome message should get displayed with the name retrieved from session.
Use HTML for developing view component and servlet as controller

## 1) WelcomeCustomer

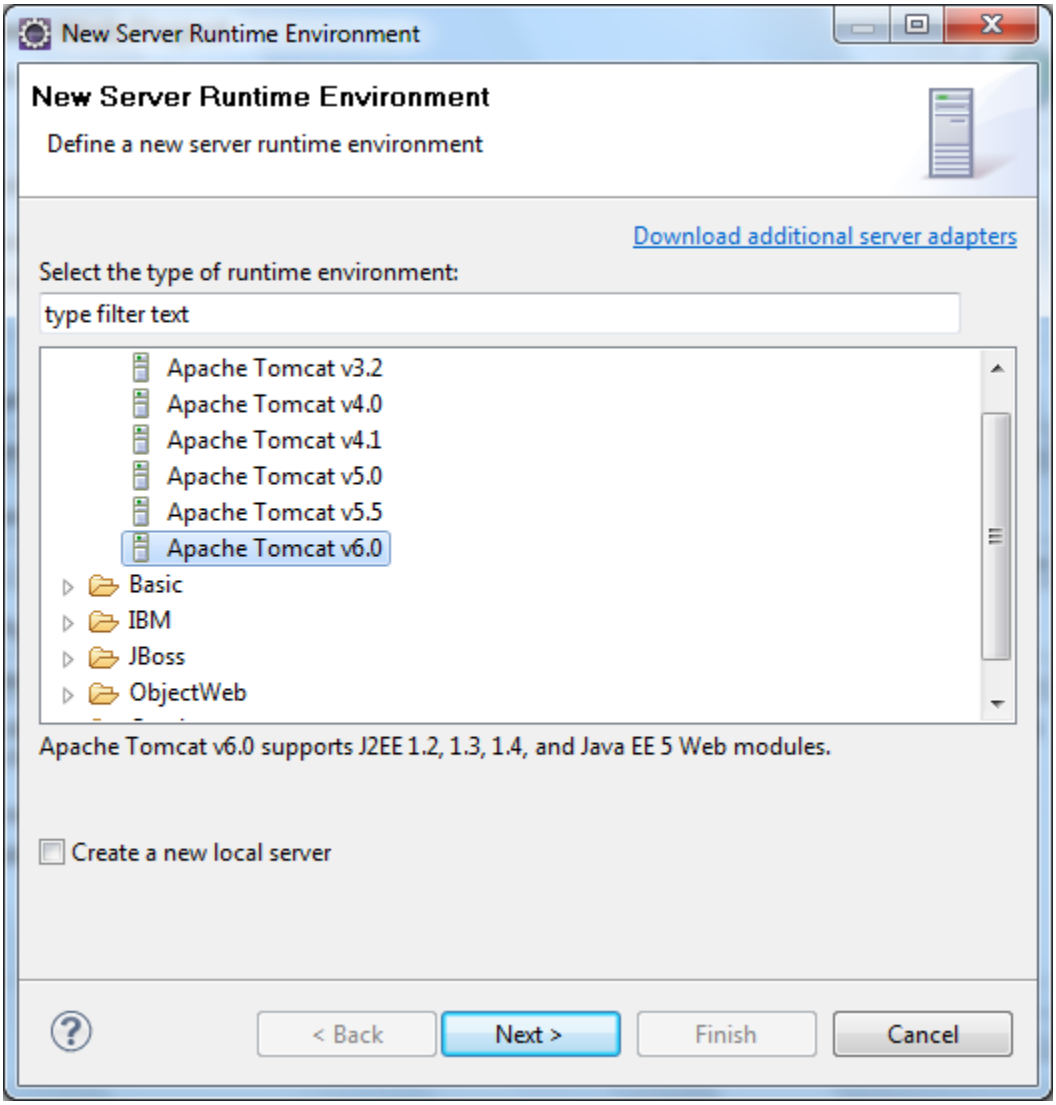
Create dynamic Web project **by File->New->Dynamic Web Project**



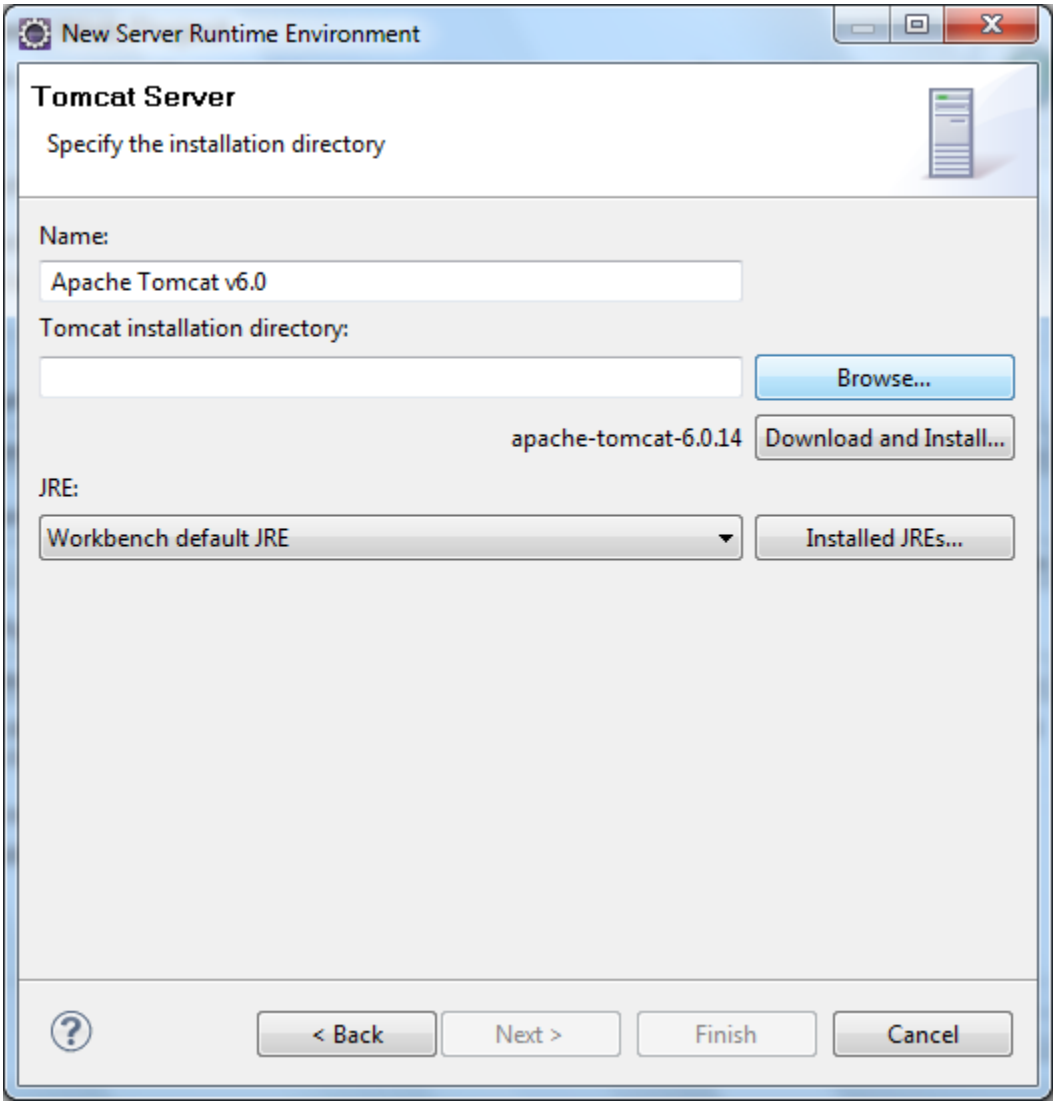
Provide a valid project name "WelcomeCustomer" adhering to the coding standards



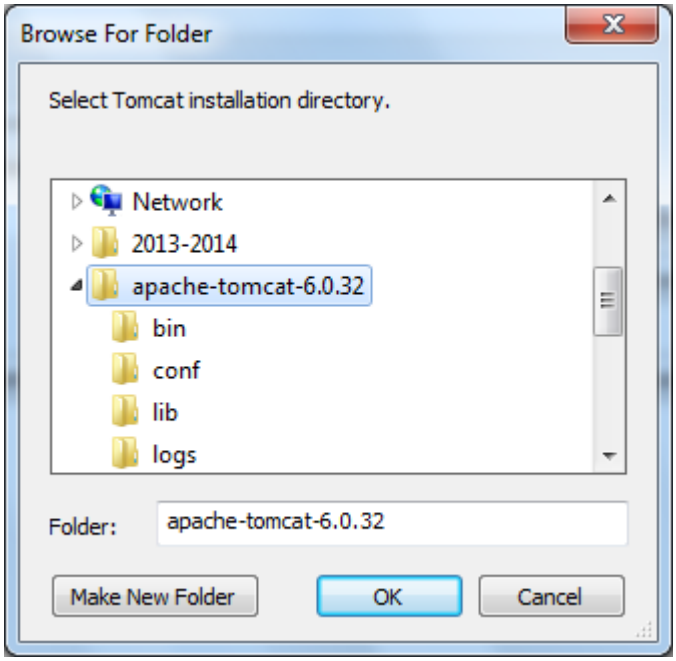
Set the new target runtime environment by clicking on New in Target runtime. Select **Apache Tomcat v6.0** and click on Next button.



Browse for the right installation directory for Apache Tomcat v6.0

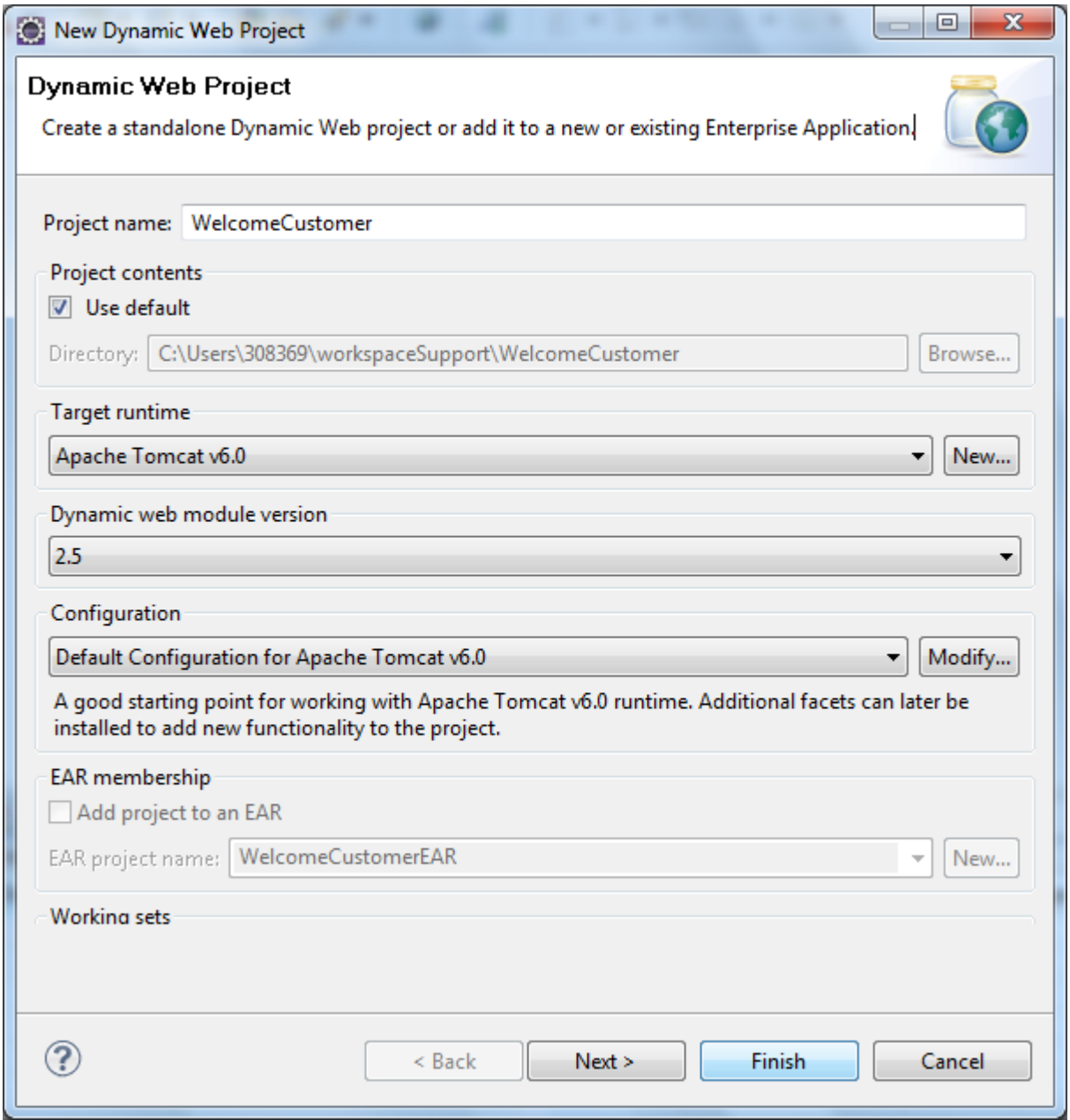


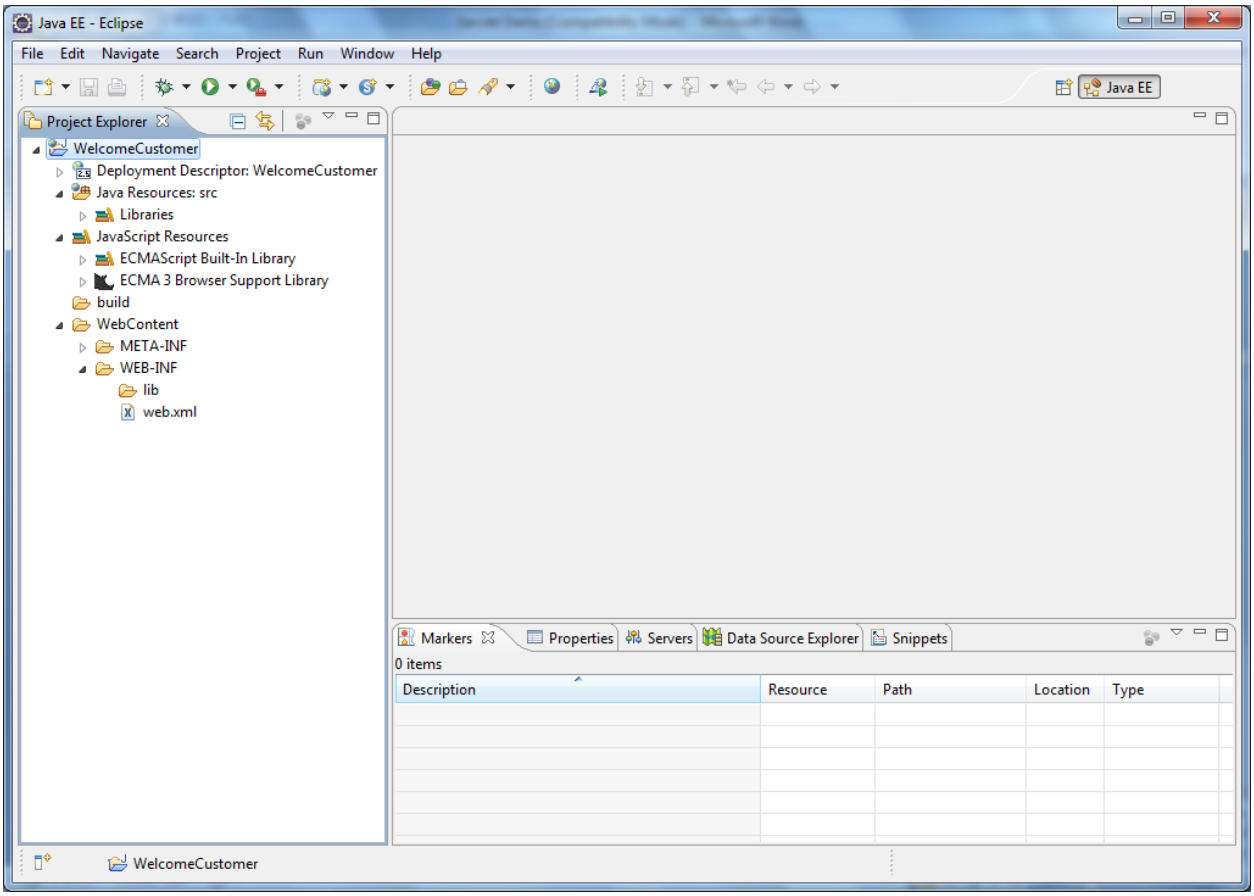
Browse for the right folder of apache-tomcat-6.032 from the local and click on OK



Once the target runtime gets populated click on finish.

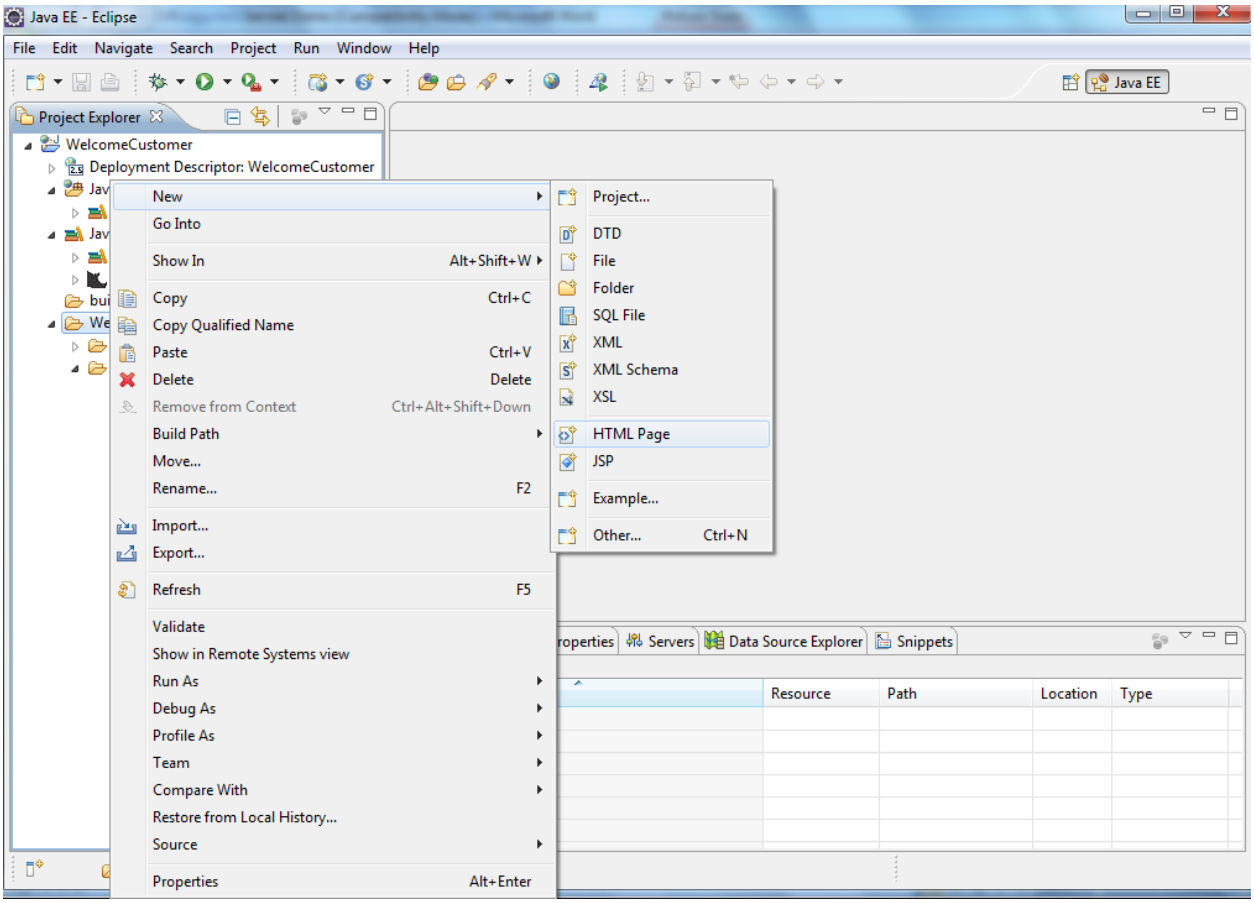
A dynamic Web Project “WelcomeCustomer” with the folder structure as follows gets created. The same can be viewed in the project explorer.



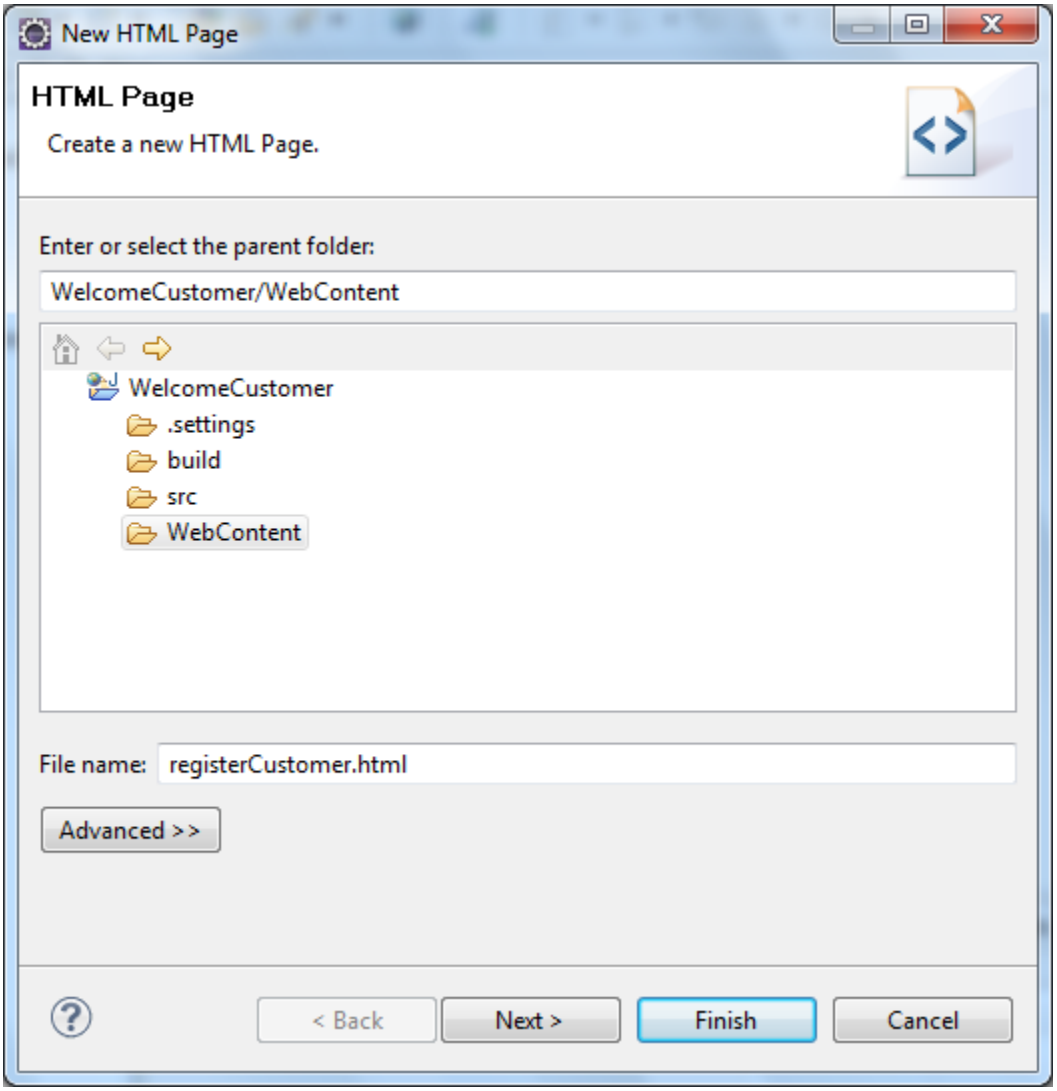



## 2) registerCustomer.html

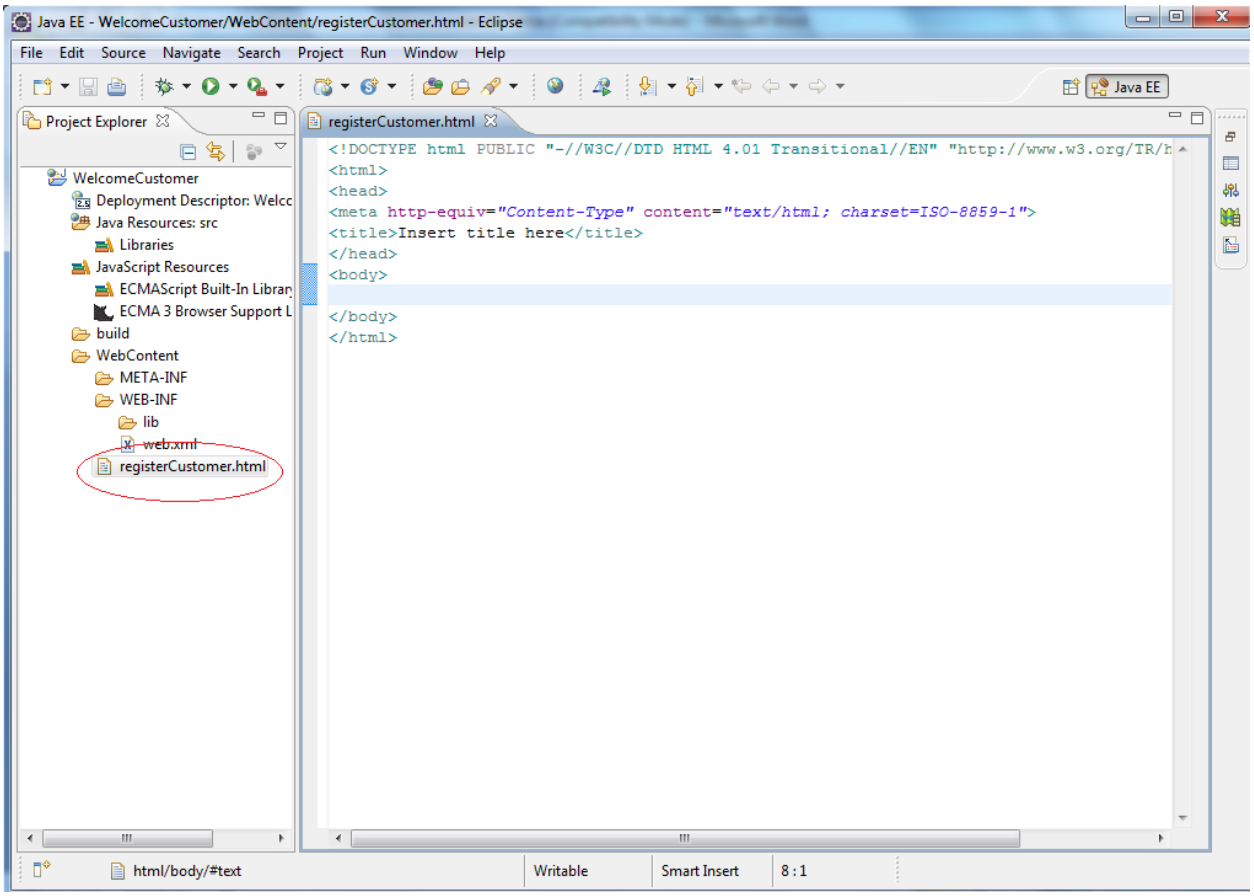
Create registerCustomer.html inside Webcontent of the Dynamic Web project WelcomeCustomer.



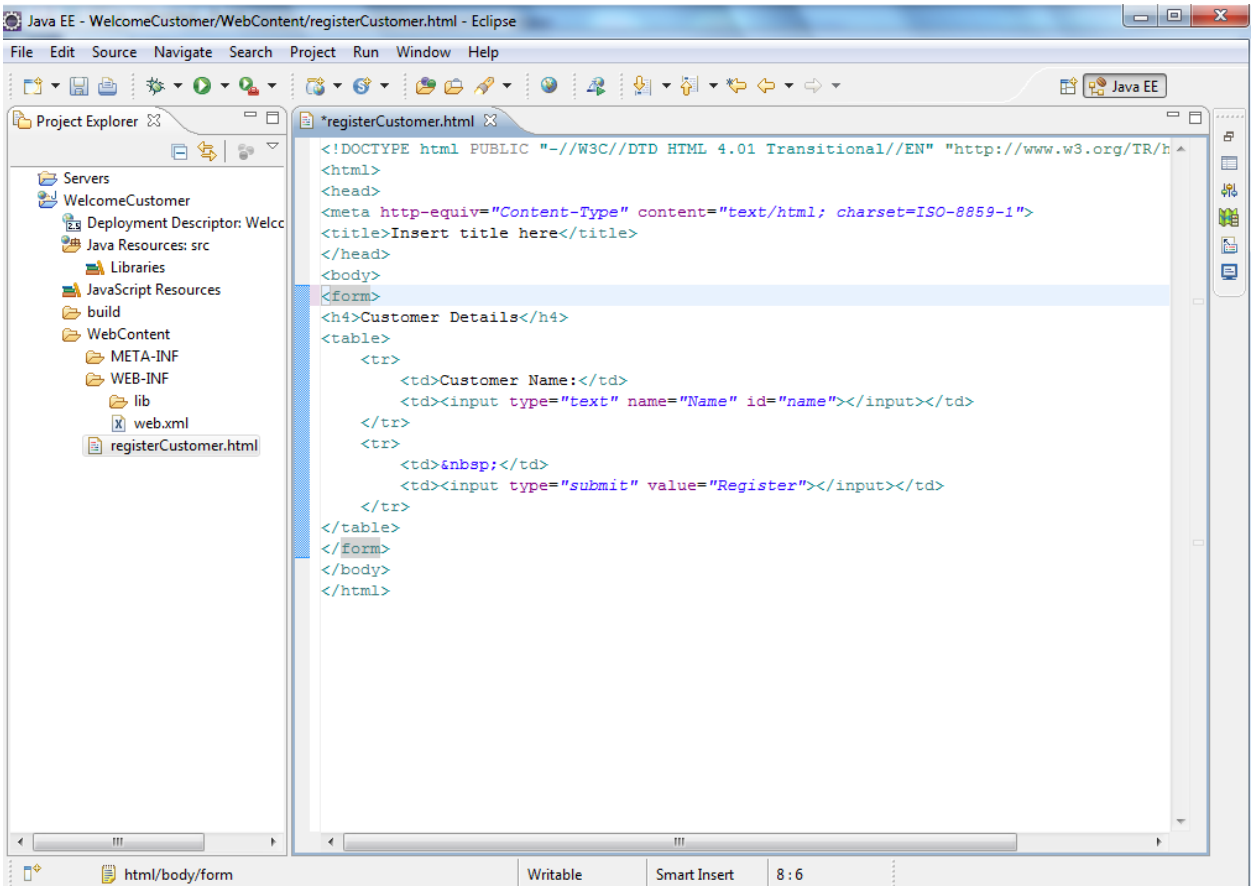
Provide a valid name for the html page and click on ‘Finish’.



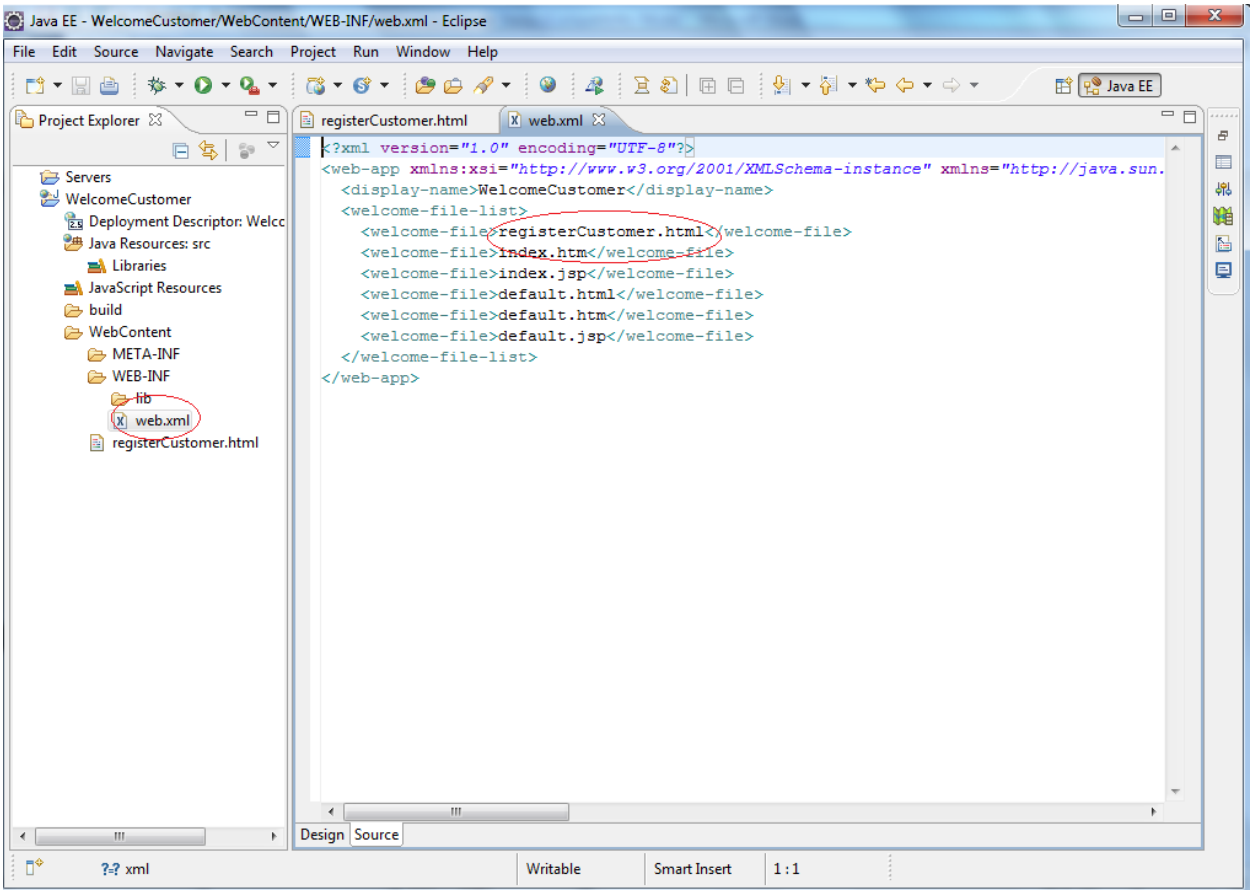
“registerCustomer.html” gets created inside the WebContent folder as highlighted



Create a form inside the registerCustomer.html that wlll get the Customer name**.**

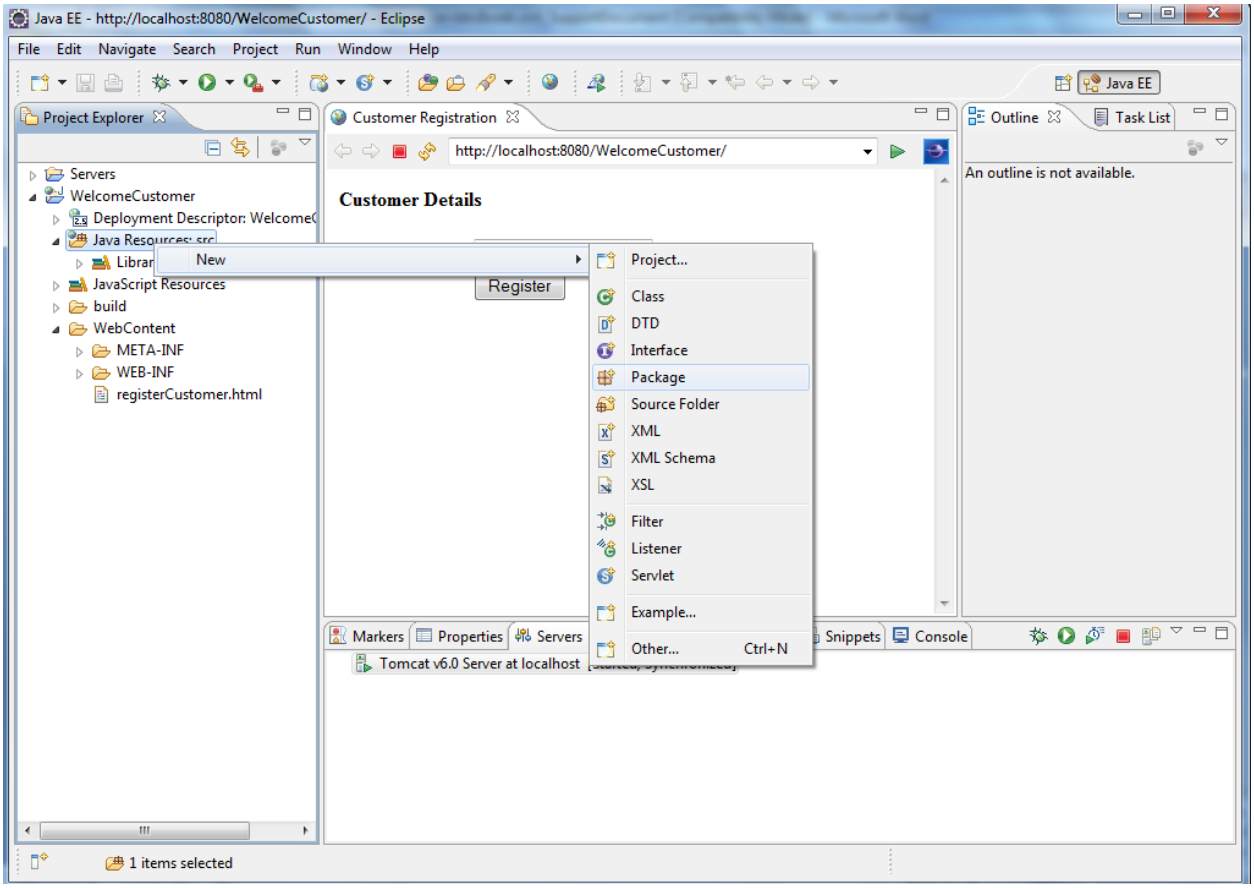


Provide the welcome file name as registerCustomer.html inside web.xml

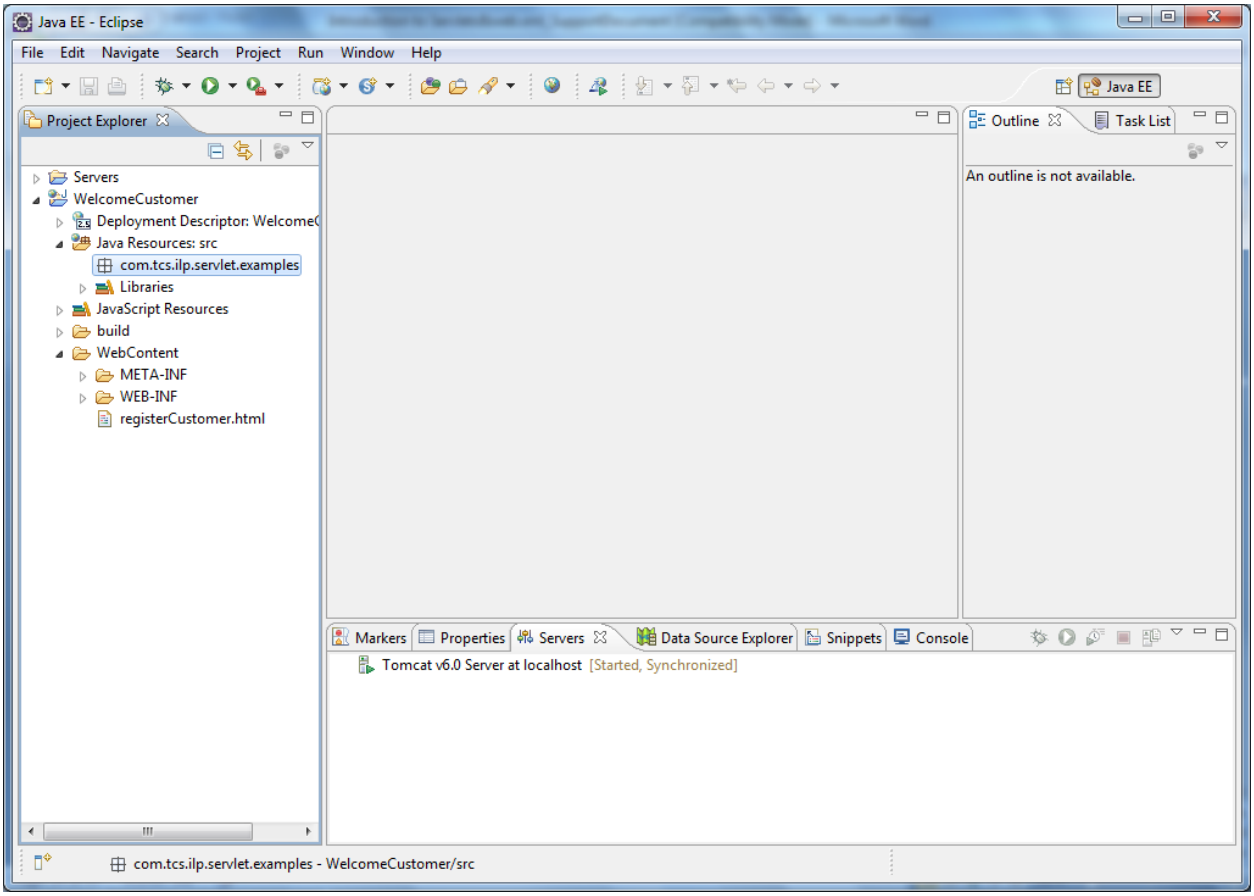


### 3) CustomerController.java

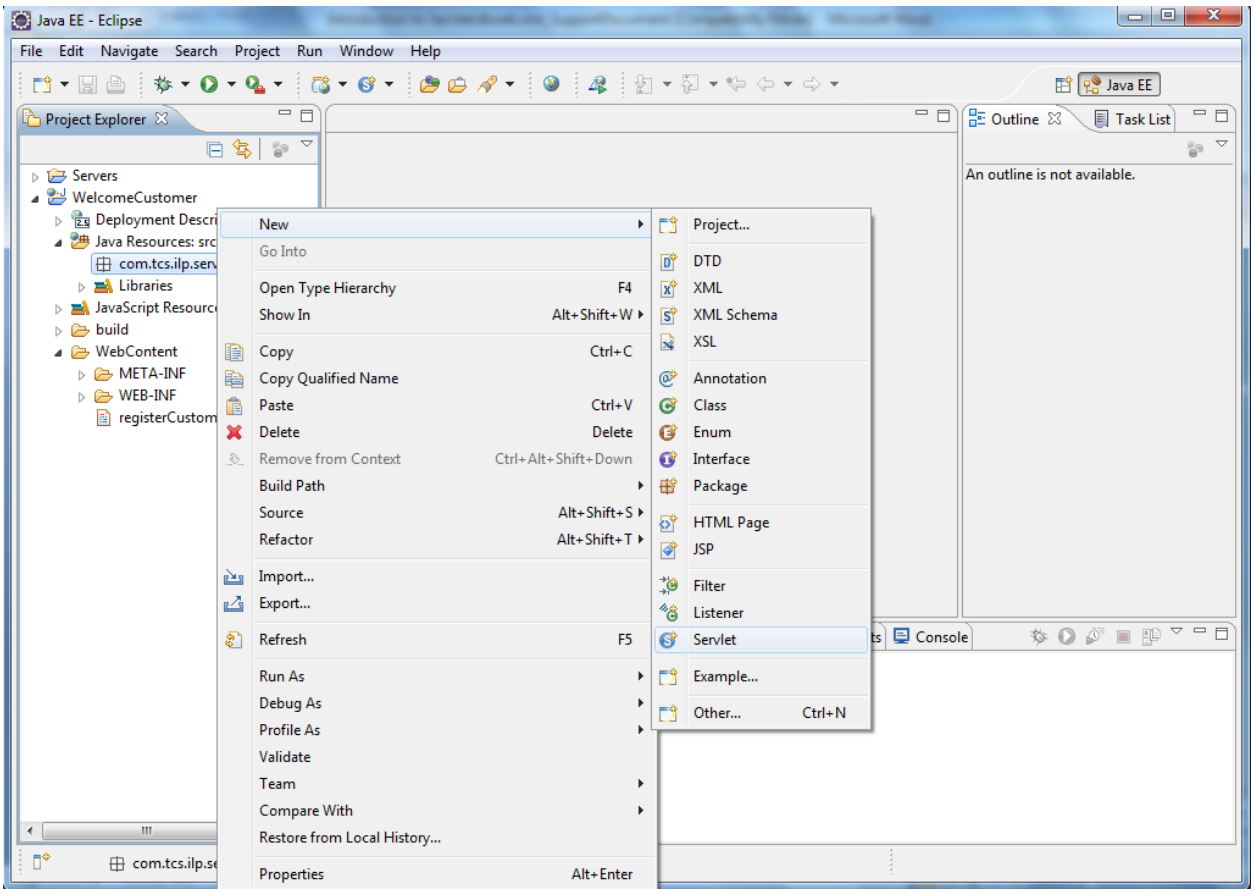
Create a package com.tcs.ilp.servlet.examples inside src folder



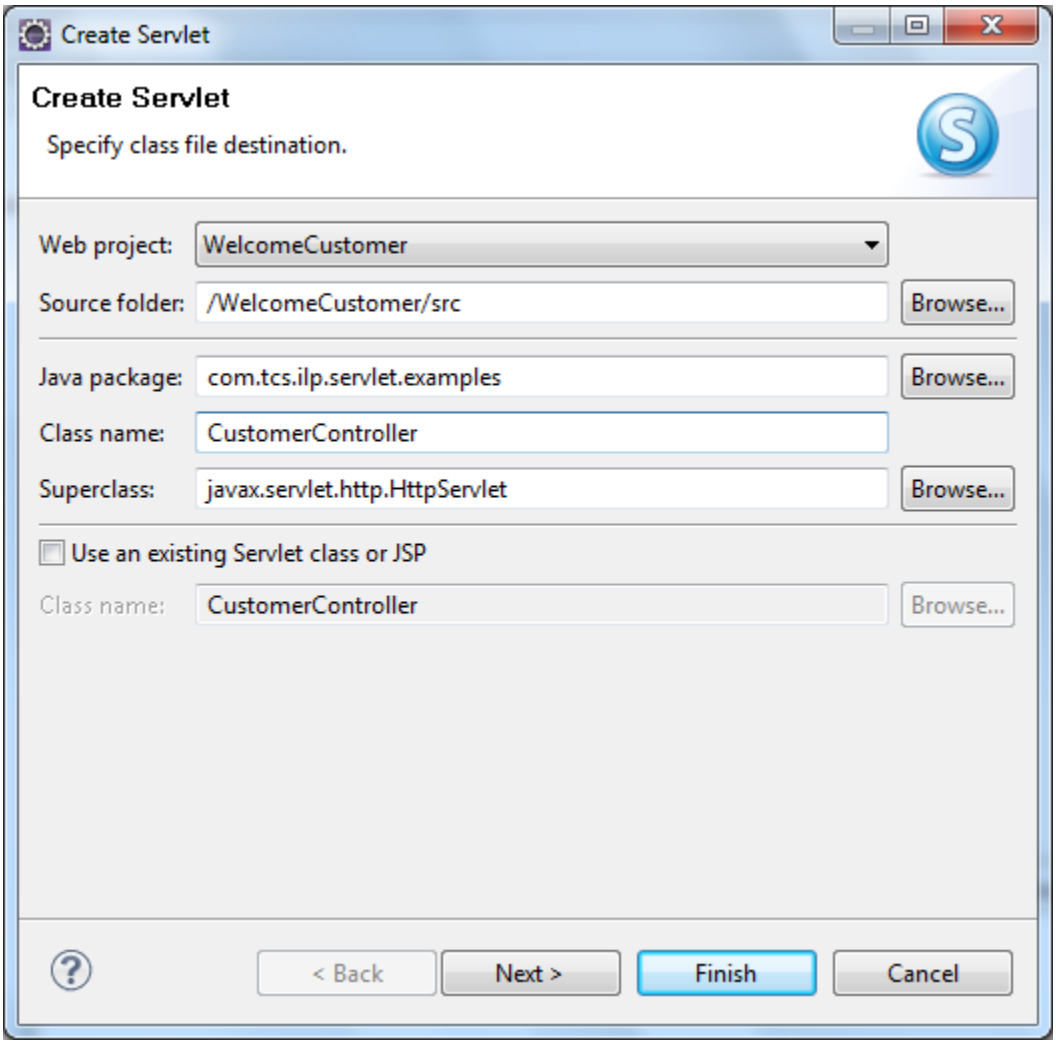
The package gets created inside the src folder



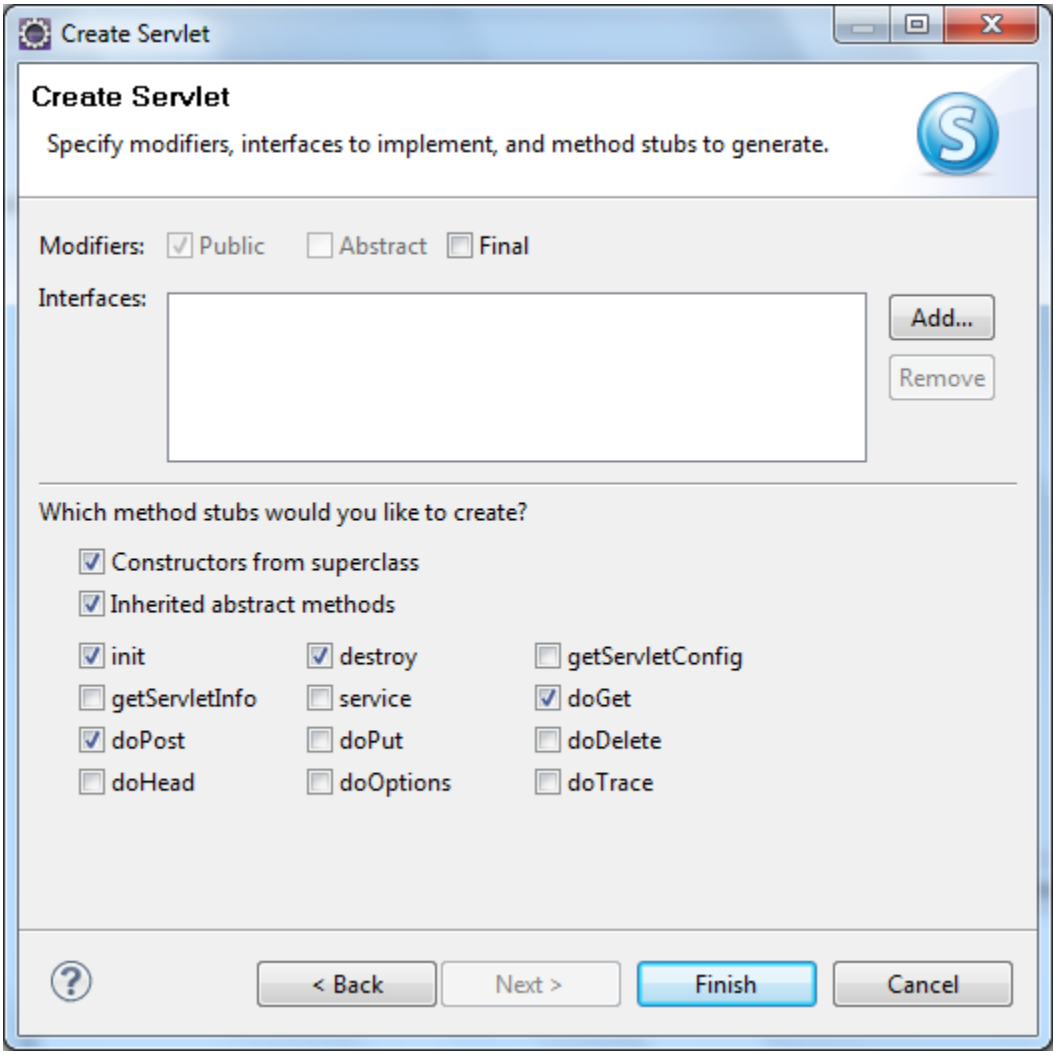
**Create a new Servlet  CustomerController inside the package.**



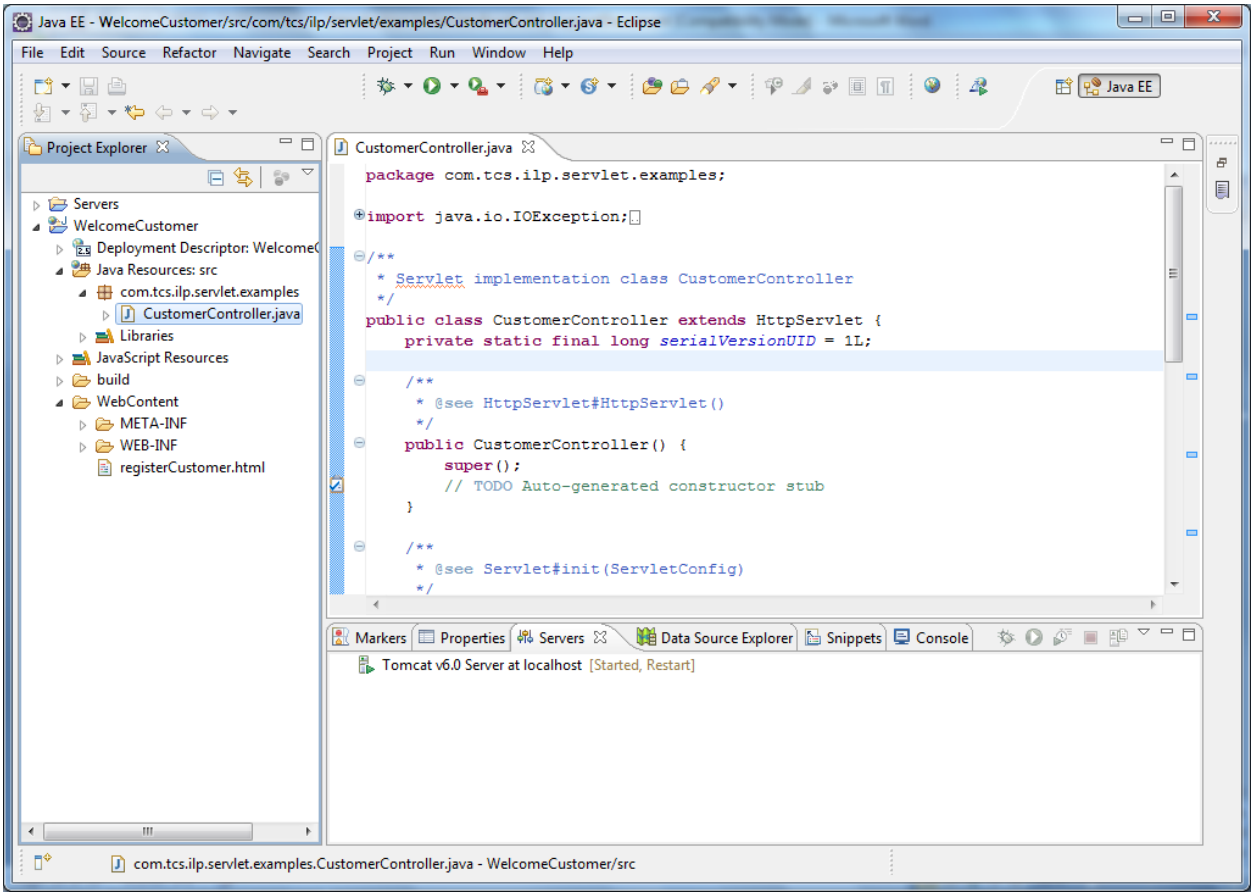
Provide a valid classname for the Servlet and click on Next



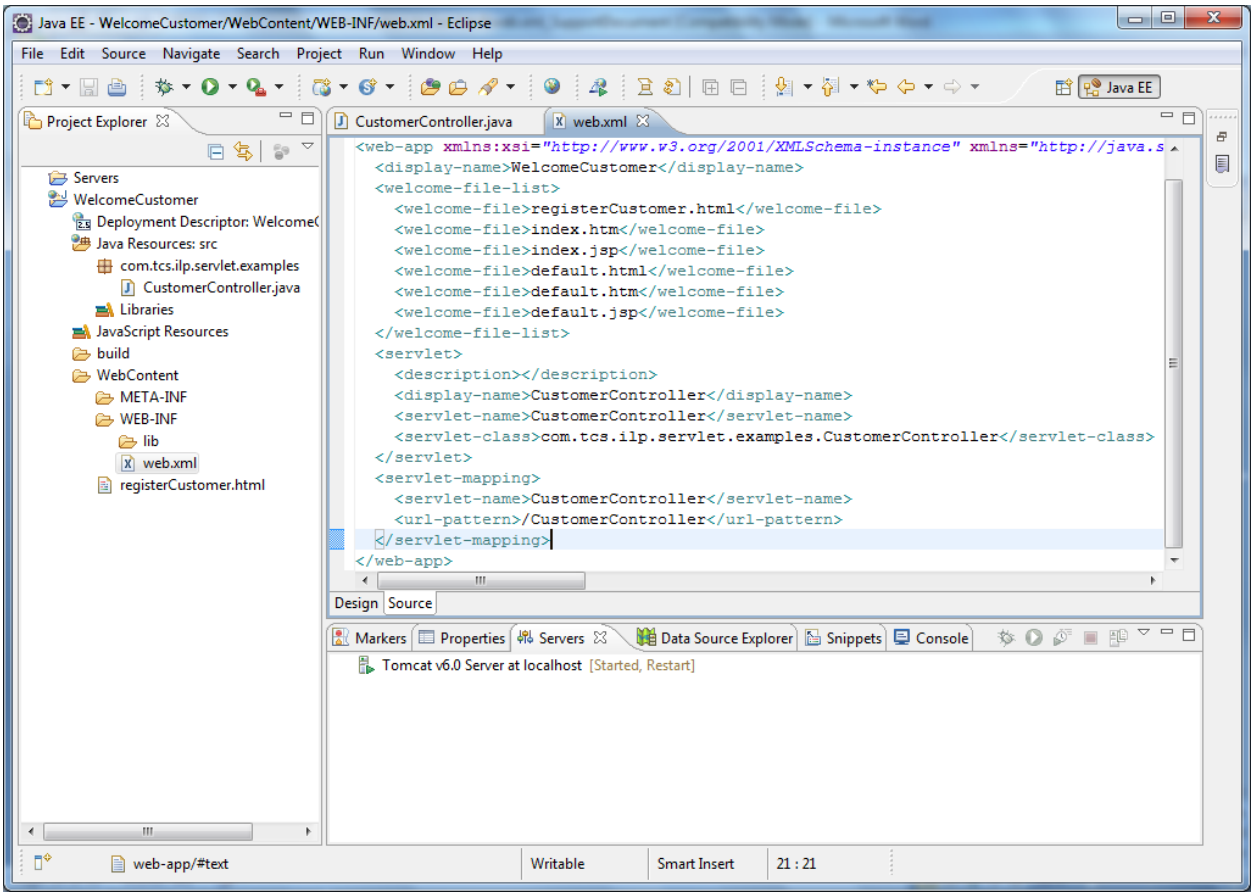
Try to override the lifecycle methods init, destroy,doGet and doPost and click on Finish



The servlet gets created under the package structure



As soon as the servlet is created the corresponding servlet mapping gets created inside the web.xml
Servlet definition has the display name, servlet name and the servlet-class it which it is available
Servlet mapping has the servlet name and the url-pattern.



**Code for retrieving the customer keyed in name through request is provided inside servlet**

Java EE - WelcomeCustomer/src/com/tcs/ilp/servlet/examples/CustomerController.java - Eclipse

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CustomerController.java

```
package com.tcs.ilp.servlet.examples;

import java.io.IOException;
import java.io.PrintWriter;

import javax.servlet.ServletConfig;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;
import javax.servlet.RequestDispatcher;

/**
 * Servlet implementation class CustomerController
 */
public class CustomerController extends HttpServlet {
    private static final long serialVersionUID = 1L;

    /**
     * @see HttpServlet#HttpServlet()
     */
    public CustomerController() {
        super();
        // TODO Auto-generated constructor stub
    }

    /**
     * @see Servlet#init(ServletConfig)
     */
    public void init(ServletConfig config) throws ServletException {
        // TODO Auto-generated method stub
```

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Java EE - WelcomeCustomer/src/com/tcs/ilp/servlet/examples/CustomerController.java - Eclipse

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CustomerController.java

```
 * @see HttpServlet#doGet(HttpServletRequest request, HttpServletResponse
 *      response)
 */
protected void doGet(HttpServletRequest request,
        HttpServletResponse response) throws ServletException, IOException {
    // TODO Auto-generated method stub
    doPost(request, response);
}

/**
 * @see HttpServlet#doPost(HttpServletRequest request, HttpServletResponse
 *      response)
 */
protected void doPost(HttpServletRequest request,
        HttpServletResponse response) throws ServletException, IOException {
    // TODO Auto-generated method stub
    String customerName = request.getParameter("Name");
    PrintWriter out = response.getWriter();

    HttpSession session = request.getSession(true); // Always Return a HTTP
    // Session Object
    session.setAttribute("User", customerName); // Setting a Session Attribute
    // object=admin
    session.setMaxInactiveInterval(15); // Setting Session to expire in 1
    // minute
    RequestDispatcher dispatcher = request
            .getRequestDispatcher("WelcomeServlet");
    dispatcher.forward(request, response);

}

}
```

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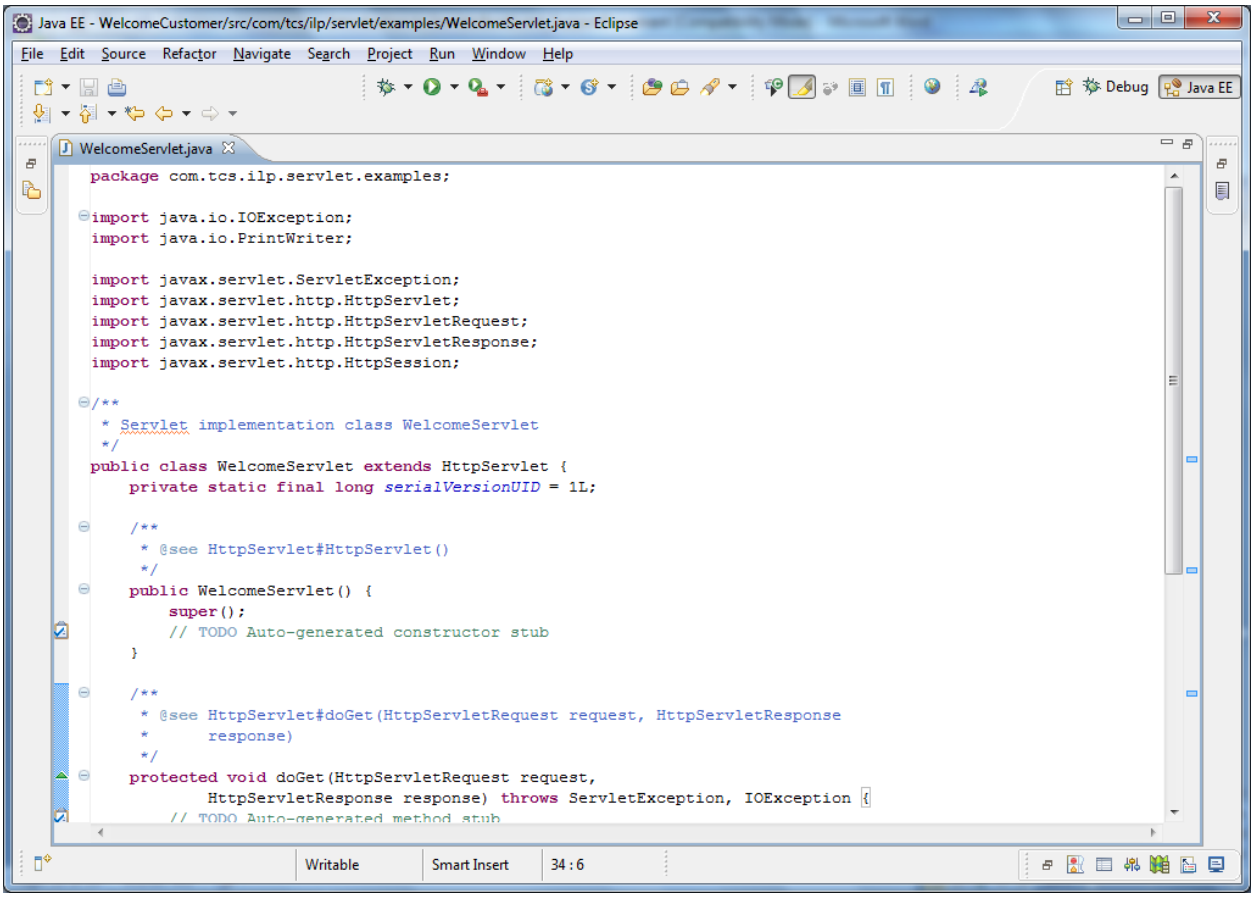
Create a Welcome Servlet that retrieve the customer name from the session and show it in the welcome message to the user



Java EE - WelcomeCustomer/src/com/tcs/ilp/servlet/examples/WelcomeServlet.java - Eclipse

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Debug Java EE

WelcomeServlet.java

```
 * @see HttpServlet#HttpServlet()
 */
public WelcomeServlet() {
    super();
    // TODO Auto-generated constructor stub
}

/**
 * @see HttpServlet#doGet(HttpServletRequest request, HttpServletResponse
 *      response)
 */
protected void doGet(HttpServletRequest request,
        HttpServletResponse response) throws ServletException, IOException {
    // TODO Auto-generated method stub
    doPost(request, response);
}

/**
 * @see HttpServlet#doPost(HttpServletRequest request, HttpServletResponse
 *      response)
 */
protected void doPost(HttpServletRequest request,
        HttpServletResponse response) throws ServletException, IOException {
    // TODO Auto-generated method stub
    PrintWriter out = response.getWriter();
    HttpSession session = request.getSession(true);
    out.println("Hello! The user session active here is: "
            + session.getAttribute("User"));
}

}
```

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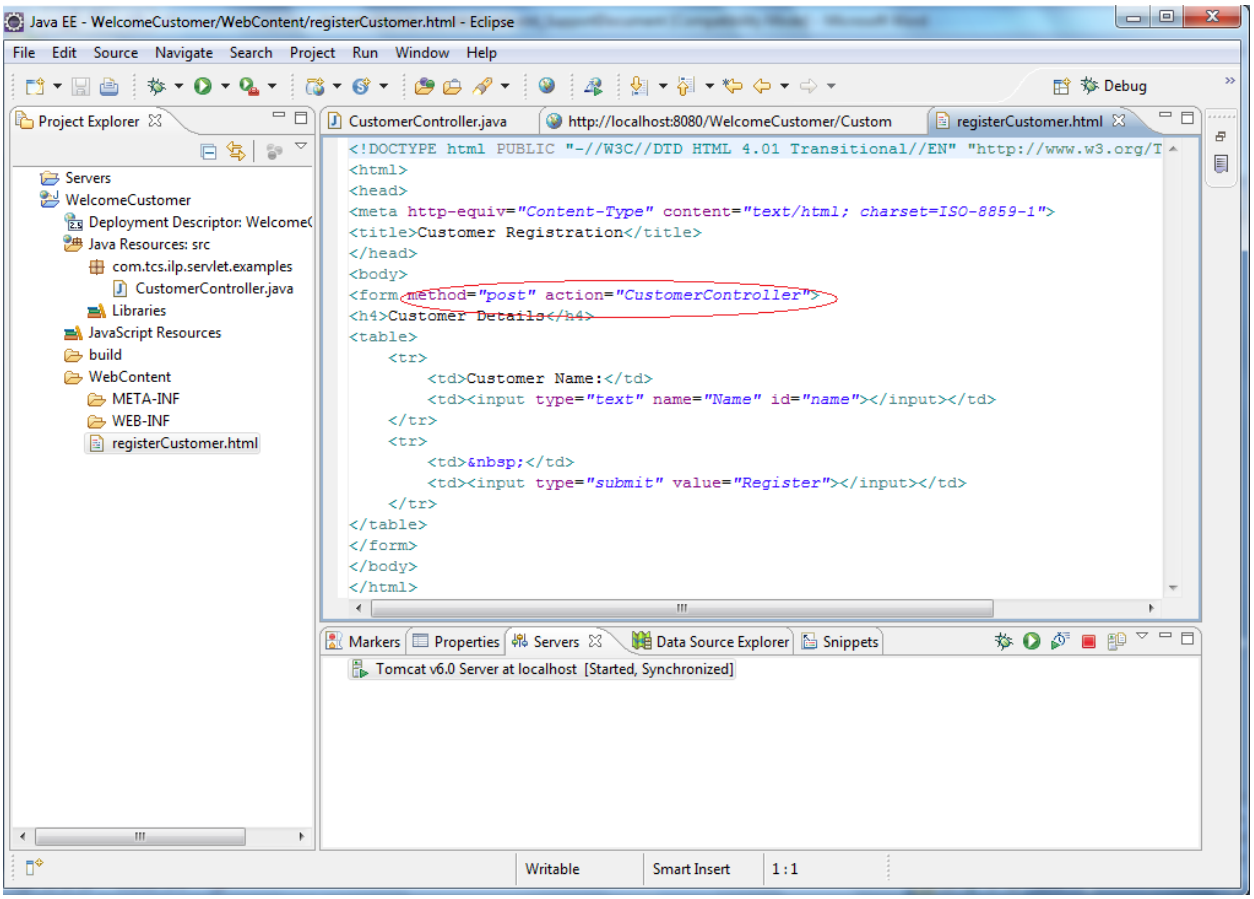
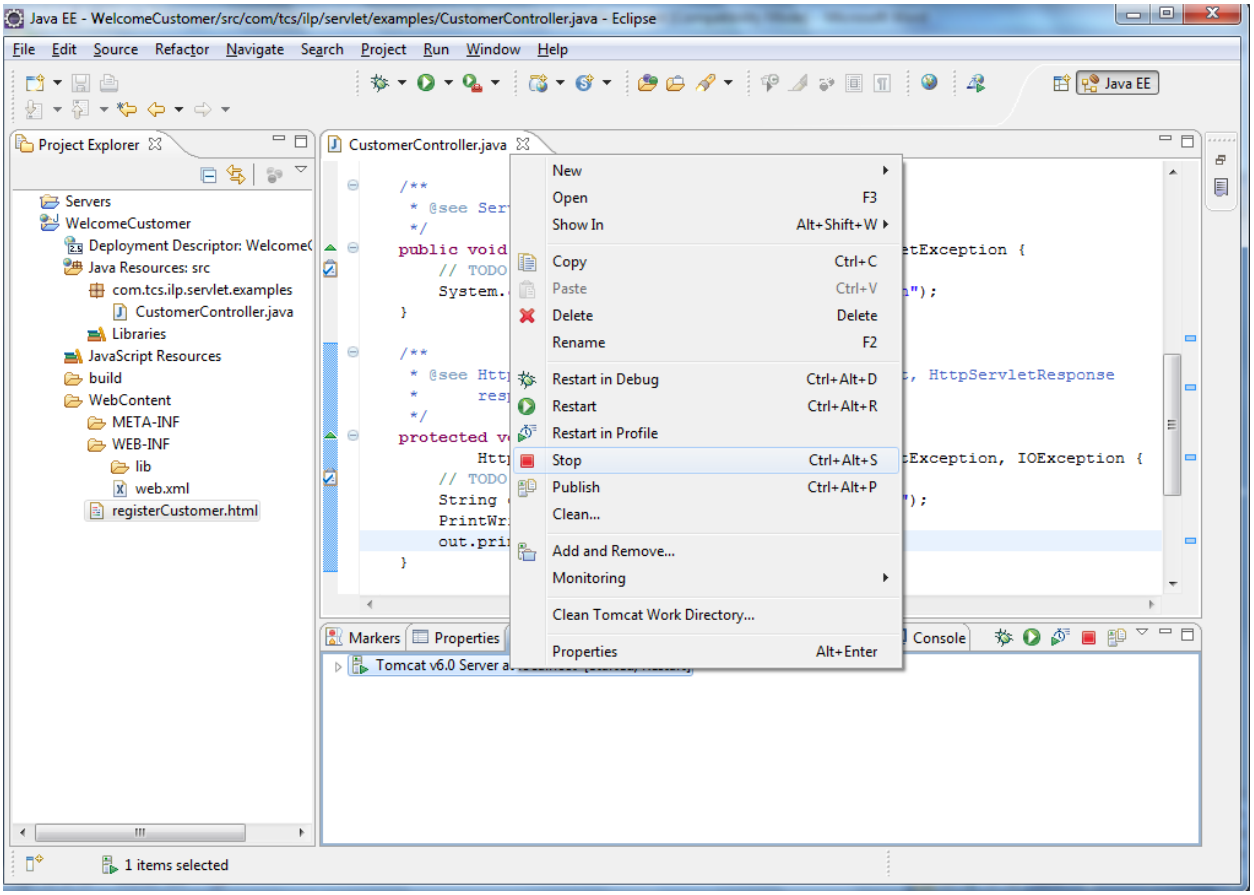
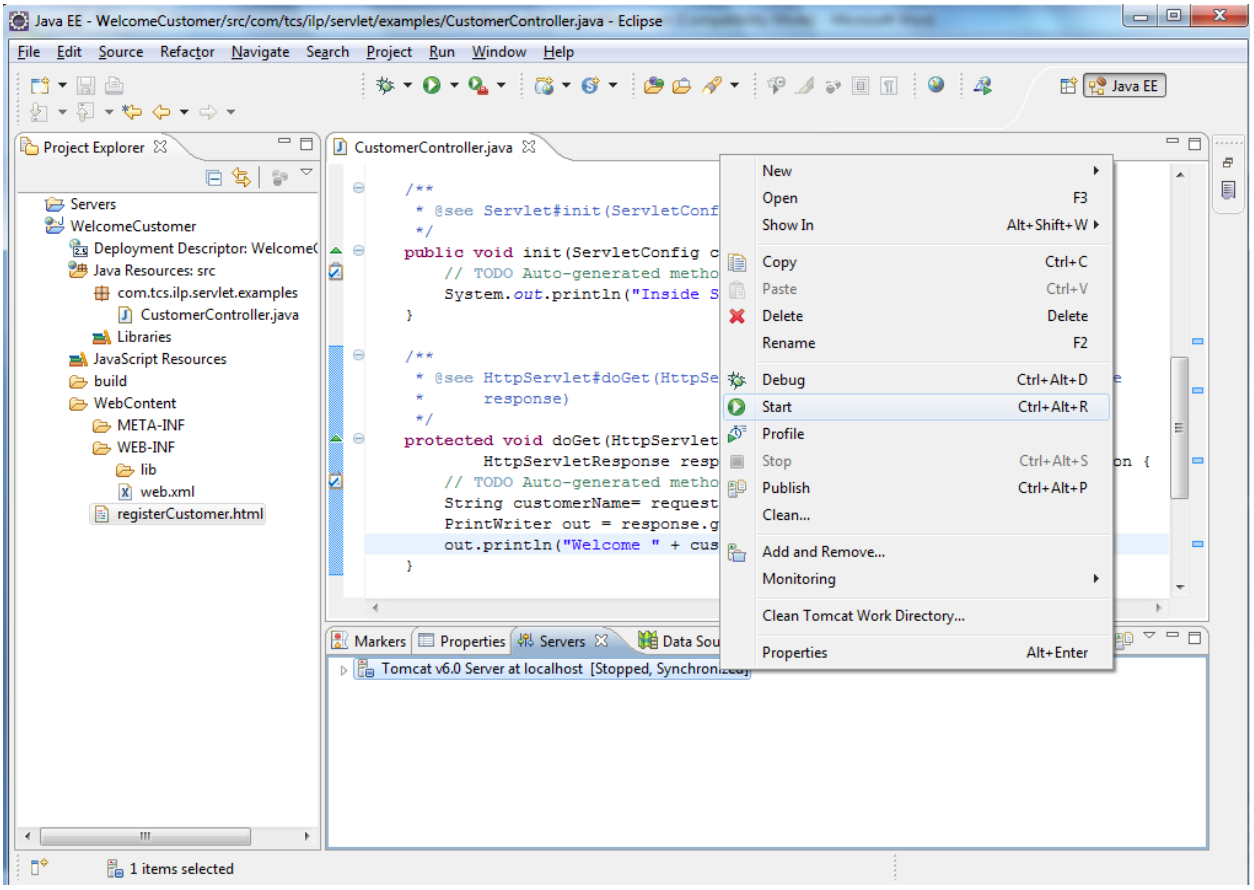
Provide the corresponding action and method inside registerCustomer.html for CustomerController



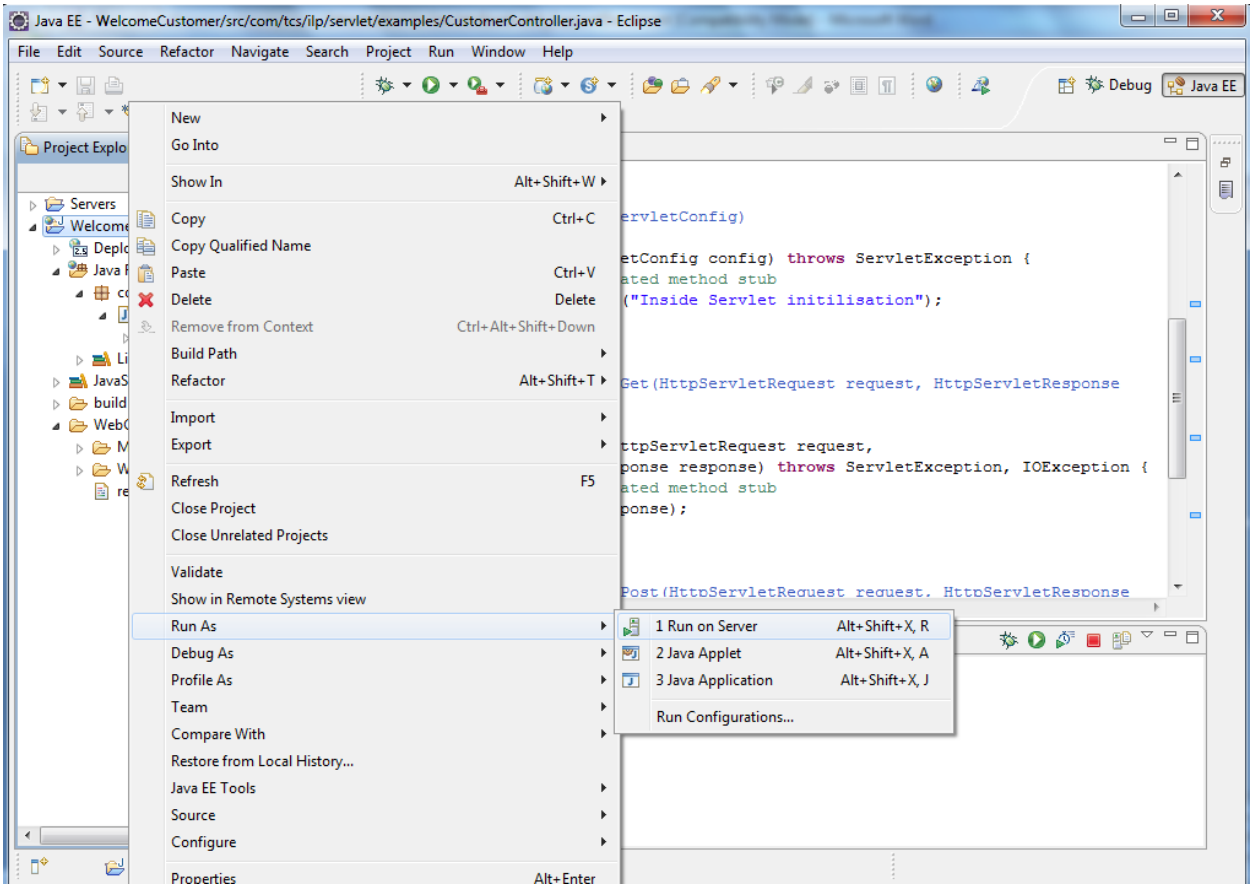
Create the server instance as in WebApplicationBasics_SupportDocument.doc.
Stop the server and restart the server again for the new changes to get reflected



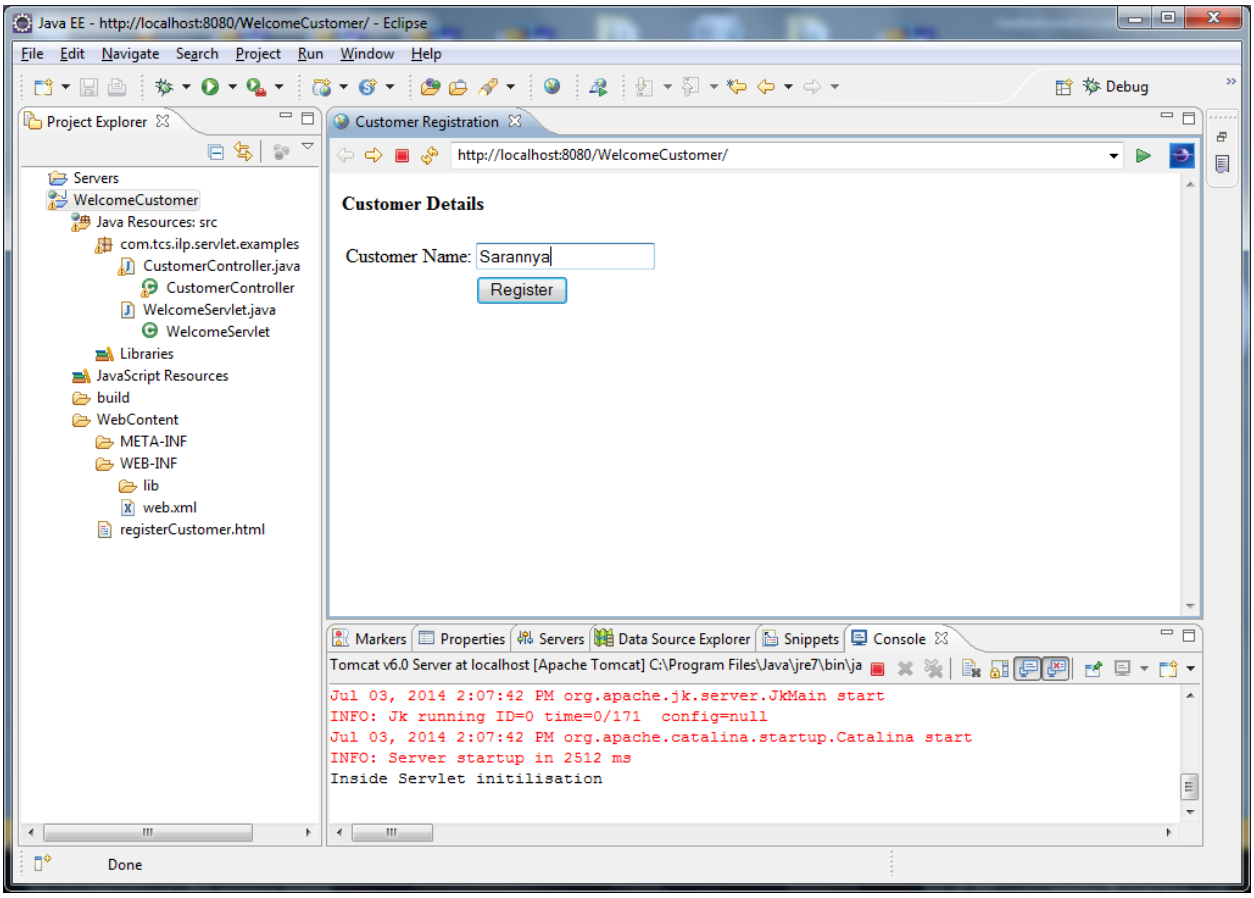
The server will be in stopped status. Now again restart the server.



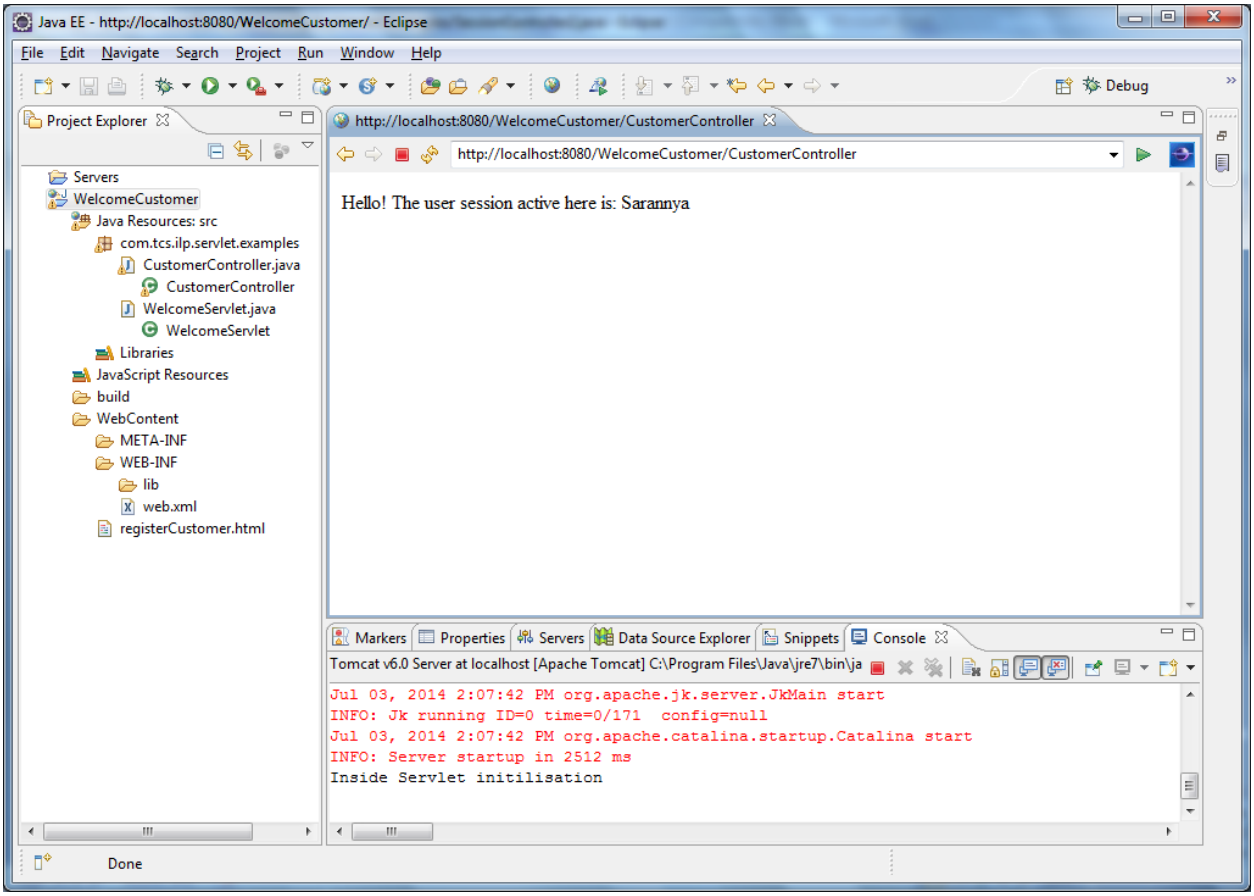
Once the server gets started, run the dynamic web project on server



The output is shown on the screen. Key in a customer name and click on register. The request gets forwarded to WelcomeServlet after storing the custromer name in HTTPSession object.



A welcome message with customer name getting displayed from session..




Note: Session management can be applied in login and logout functionality. For details follow the link http://www.javatpoint.com/servlet-http-session-login-and-logout-example